Arjun Mauji

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Professional Summary

Mechanical Engineering undergraduate at Queen's University with hands-on experience in mechanical and embedded systems design, along with strong problem-solving and communication skills developed through collaborative engineering projects.

Education

B.A.Sc. Mechanical Engineering, Queen's University *Kingston, Canada | 2027*

- Dean's Scholar 24/25, Biggs – Ronald & Deanna Scholarship
- Relevant Coursework/Concepts: Embedded Programming, Real Time Operating Systems, Microcontrollers, Fluid Dynamics, Thermodynamics, Material Science

Work Experience

Teaching Assistant (Mechatronics Course), Queen's University Sep – Dec 2025

- Co-supervised 10 mechatronics labs with over 100 students on measurement sensors and signal noise filtering
- Answered electronic measurement theory questions and troubleshot lab circuits

Projects

Digital Oscilloscope, Personal Project | GitHub July 2026

- Designing a digital **STM32**-based oscilloscope to sample and visualize signals at 1 Msps in the 0-3.3 V range
- Implementing a high impedance front end, tested in **LTSpice**, to safely probe signals up to 100kHz
- Using **DMA** to enable continuous signal sampling via the board's ADC and minimize CPU workload
- Developing the firmware in **bare-metal C**, with a custom SPI driver and build process

Embedded Team, Queen's Off-Road Car Design Team (Baja) Feb – Apr 2026

- Programmed **STM32** boards in C to monitor the vehicle transmission system's thermal condition
- Assisted in migrating the vehicle's existing **CAN** bus communication library to use **FreeRTOS**

Two Stage Shifting Gearbox, Team Project Jan – Apr 2026

- Built a 2-stage shifting gearbox and housing to balance torque and speed across an RC car's climb and sprint modes
- Performed **AGMA fatigue analysis** and iterated gear geometry to achieve fatigue safety factors above 1.0
- Adapted an initial steel design for 3D printing and modeled the gearbox assembly in **SolidWorks**

Hardware Motion Analyzer, Personal Project | GitHub Nov – Dec 2025

- Prototyped a low-cost (<$40) wireless motion analyzer to compare dynamic movement flexibility between users
- Programmed 2 IMU-based sensor modules to capture movements by calculating joint flexion and extension angles
- Implemented RTOS tasking with **I2C** communication to log and filter up to 10 angle data streams concurrently
- Analyzed movement flexibility and peak angle values using **SciPy**

Skills

Mechanical: SolidWorks CAD & FEA, Reading Assembly Drawings, Ansys (Fluent)

Embedded Systems: STM32, ESP32, FreeRTOS, CAN, SPI, UART

Software Tools: C, Python, MATLAB (Simulink), Git/GitHub, GDB

Electrical Tools: Multimeters, Soldering, Oscilloscopes, LTSpice